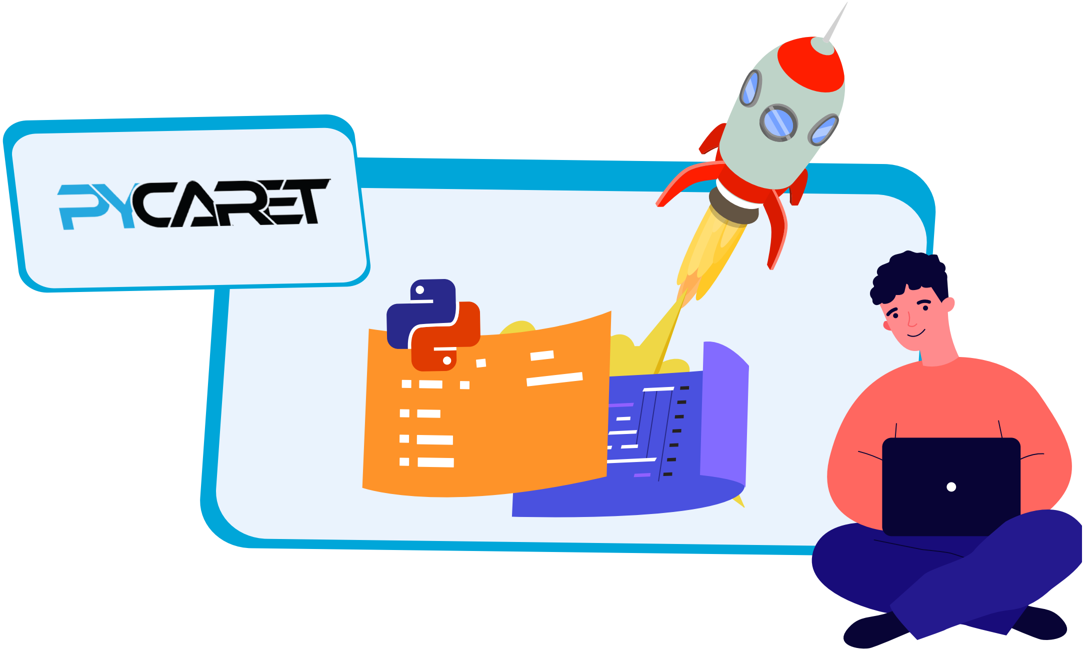
Automating Machine Learning with PyCaret

Learn how to use PyCaret, a low-code Python library, to automate the management of machine learning models.



Machine learning (ML) and other data science technologies are today being used in almost every field. A basic awareness about these technologies is therefore important. However, developers may not be interested in or be good at developing or writing large pieces of code. Tools that help us automate ML processes are of immense help in this scenario. One such open source tool is PyCaret.

PyCaret is a low-code Python library that helps in end-to-end ML and model management. It not only helps novice coders but also saves a lot of time for experienced machine learning developers. It can be used to simplify the training, testing and tuning of ML models. PyCaret can replace hundreds of lines of code written using other libraries with just a few commands. It is wrapped around many popular ML libraries and tools like XGBoost, scikit-learn, Spacy, and LightGBM, amongst others. With the advent of such tools, data scientists without deep knowledge in AI and ML can also develop effective machine learning models for a wide range of use cases.

Let us look at the working and major functionalities of PyCaret. You can use any IDE for this purpose. I will be using Google Colab, which is the easiest for a beginner because you don't have the extra burden of setting up an environment on your local system. You can just open the link at *https://colab.research.google.com/* and create a new notebook there. Once you do that, click on the 'Connect' button. This will connect your notebook to a kernel, which will help you run your code.

The first step is to install the library. This can be done using the following command:

```
!pip install pycaret
```

You can type this command and then press 'Shift' and 'Enter' together to run the cell.

The next step is to load a dataset. The library already comes with a lot of standard datasets. Let us explore a diabetes prediction dataset. We can load this dataset using the following command as shown in Figure 1.

```
import pycaret
from pycaret.datasets import get_data
```